SCIENCE EXPLORER / GRADE 8

The Cell

Watch. Predict. Explain.

Original revision guide with 100 practice questions and answer explanations.

Original teaching resource informed by KICD's revised 2024 Grade 8 curriculum. Five introductory missions for strand 2.0, sub-strand 2.1 The Cell. The curriculum allocates 20 lessons to this sub-strand; this is not a full replacement. Independent prototype, not KICD-endorsed. Teacher review required. The source is linked, never bundled.

KICD Integrated Science Grade 8, revised 2024. strand 2.0, sub-strand 2.1 The Cell. Source viewer pages 21-22 of 37. Newly written explanations, questions and visuals, not an official assessment or a copy of the source.

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Virtual models only. Real microscopes, glass, stains and biological specimens require a teacher-approved method. Do not collect mouth samples, use chemicals or handle broken glass unsupervised.

Use the HTML animation first, try its prediction, then read the optional explanation. All 100 practice questions are below. Short answers are self-assessed; equivalent scientifically correct wording is acceptable. No personal health information should be entered.

Your learning route

1. Meet the cell: Recognise cells as basic units of living things and compare simple models.

2. Parts with jobs: Connect visible cell structures with their functions.

3. Compare fairly: Use shared features and differences without overgeneralising.

4. Virtual microscope: Determine total magnification and recognise its limits.

5. Observe, draw, care: Distinguish observations from assumptions and care for practical equipment.

1. Meet the cell

Recognise cells as basic units of living things and compare simple models.

Cells are basic structural and functional units of living things. Some organisms consist of one cell; others contain many cells with different roles. A cell is not the same thing as an atom.

A typical plant-cell model includes a cell wall, cell membrane, cytoplasm, nucleus and a large central vacuole. Photosynthetic plant cells can contain chloroplasts. A typical animal-cell model has a membrane, cytoplasm and nucleus but no cell wall or chloroplasts.

These diagrams are simplified, not to scale and not universal for every cell. Not every plant cell contains chloroplasts, and specialised cells may differ from the typical models.

Light microscopes reveal features depending on the specimen, preparation and instrument. A coloured diagram is easier to label than a real specimen and should not be mistaken for an exact microscope view.

Visual mission

Two cells. Shared essentials.

Typical plant and animal cells share a membrane, cytoplasm and nucleus.

Typical diagrams, not to scale. Not all plant cells contain chloroplasts.

Think it through

Name two structures shared by the typical plant and animal models and one difference.

2. Parts with jobs

Connect visible cell structures with their functions.

The cell membrane regulates movement of substances into and out of the cell. It is not simply an open hole. The cytoplasm is the material where many cell reactions occur.

The nucleus contains genetic material and helps control cell activities. It is not the cell wall, and it does not mean every specialised cell must have the same appearance.

The plant cell wall supports and helps maintain shape. It is outside the cell membrane. A large central vacuole contains cell sap and contributes to support when the cell is turgid.

Chloroplasts in photosynthetic plant cells carry out photosynthesis. A light microscope cannot reliably reveal every tiny structure shown in an advanced cell diagram; focus on the features the specimen and instrument support.

Visual mission

Tap a part. Discover its job.

A label becomes useful when you can explain the function.

Colours and sizes are educational. A light microscope may not show every drawn detail.

Think it through

Choose a structure, describe its job and explain one mistake that could confuse it with another part.

3. Compare fairly

Use shared features and differences without overgeneralising.

Compare like with like: use labelled typical plant and animal models, then distinguish their shared features from their differences. Both models have membranes, cytoplasm and nuclei.

The illustrated photosynthetic plant cell also has a wall, chloroplasts and a large central vacuole. The illustrated animal cell lacks a plant wall and chloroplasts and does not have the same large central vacuole.

A regular box-like drawing can help show a wall; a rounded drawing can help show a flexible animal-cell outline. Shape alone is not a reliable universal test for cell type.

A cell from an onion bulb may show a wall and nucleus without chloroplasts. An observation should not be declared wrong merely because it lacks every feature of a green-leaf diagram.

Visual mission

Typical does not mean every

Cell type and function matter more than diagram colour.

Use the example selector. Model colours are not literal specimen colours.

Think it through

Explain why 'all plant cells are green boxes' is not a reliable scientific statement.

4. Virtual microscope

Determine total magnification and recognise its limits.

A compound light microscope uses an eyepiece lens and an objective lens. Total magnification is the eyepiece magnification multiplied by the objective magnification.

For example, a 10-times eyepiece with a 10-times objective gives 100-times total magnification; with a 40-times objective it gives 400-times total magnification. This is multiplication, not addition.

Magnification makes an image appear larger. Resolution concerns distinguishing nearby details. Enlarging a blurred image does not automatically reveal new real detail.

Use a real microscope only as taught. Start and focus using the teacher's approved method, handle glass safely and do not force a lens towards a slide. The virtual lens selector requires no real equipment.

Visual mission

Turn the virtual lens

Total magnification multiplies the eyepiece and objective values.

Illustrative field of view, not measured cell size. Larger does not automatically mean more resolved detail.

Think it through

Use the virtual selector to compare 100-times and 400-times views. Explain why a larger image is not automatically sharper.

5. Observe, draw, care

Distinguish observations from assumptions and care for practical equipment.

A useful biological drawing communicates what was observed. Use clear lines, an appropriate title and accurate labels following the teacher's conventions. Do not copy every feature of a textbook as if it was visible.

A temporary slide is prepared for short-term observation. A teacher must approve the specimen, handling method, liquids and any stain. Stains and glass can pose hazards; the app gives no unsupervised slide-making procedure.

A coverslip helps form a suitable viewing preparation, but bubbles and debris can create artefacts. A dot in a field of view is not automatically a cell or nucleus.

Care for the microscope and dispose of materials as instructed. Do not use mouth cells, sharp glass, chemicals or collected biological samples without an approved supervised method. The virtual focus model is an illustration, not a real observation.

Visual mission

Look carefully. Record honestly.

A clear drawing records evidence, not every textbook feature.

This is a generated illustration, not a real microscopy result or practical slide-making guide.

Think it through

Describe how you would label a feature honestly if you could not identify it confidently.

Revision cards

Cell

A basic structural and functional unit of living things.

Membrane

Regulates entry and exit.

Cytoplasm

Where many cellular reactions occur.

Nucleus

Contains genetic material and helps control activities.

Cell wall

Plant support outside the membrane.

Chloroplast

Carries out photosynthesis in photosynthetic plant cells.

Central vacuole

Contains cell sap and helps support a turgid plant cell.

Magnification

Eyepiece multiplied by objective.

Resolution

Ability to distinguish nearby details.

Observation

Record what is visible, including uncertainty.

100-question practice bank

40 multiple-choice and 60 short-answer questions. This is a practice bank, not a standardised exam.

Meet the cell

1. A cell is best described here as what?

A. A chemical element

B. A fire class

C. A unit of pressure

D. A basic unit of living things

2. Which structure surrounds a typical animal cell?

A. A metal shell

B. A microscope lens

C. Cell membrane

D. Plant cell wall

3. Which feature is present in the typical plant model but not the animal model?

A. Nucleus

B. Cell wall

C. Cytoplasm

D. Cell membrane

4. Which statement about plant cells is accurate?

A. Not every plant cell has chloroplasts

B. All contain identical numbers of chloroplasts

C. None has a membrane

D. All are the same size

5. Are the diagram's colours literal microscope colours?

A. Always

B. Only the green background

C. They measure pressure

D. Not necessarily

6. Which organisms may consist of a single cell?

A. All trees

B. No living organisms

C. Some living organisms

D. All animals

7. Which pair is shared in typical plant and animal cells?

A. Fuel and oxygen

B. Membrane and cytoplasm

C. Cell wall and chloroplasts

D. Steel and copper

8. Why is a model not a universal picture of every cell?

A. Cells vary and can be specialised

B. No cells exist

C. Every cell is drawn to scale

D. All diagrams are measurements

9. Define a cell in this lesson.

10. Name two parts shared by the typical cell models.

11. Name one difference between the typical plant and animal models.

12. Does every plant cell contain chloroplasts?

13. What does unicellular mean?

14. What does multicellular mean?

15. Why is a cell diagram not necessarily drawn to scale?

16. Why can a microscope specimen look different from a textbook diagram?

17. Is a cell the same thing as an atom?

18. Give a reason cells can have different shapes.

19. Which typical model has a rigid outer wall?

20. Why should you read a diagram's key?

Parts with jobs

21. Which structure regulates entry and exit of substances?

A. Cell wall only

B. Microscope stage

C. Eyepiece

D. Cell membrane

22. Where do many cell reactions occur?

A. In the slide label

B. Only in the cell wall

C. Cytoplasm

D. Outside the objective lens

23. Which structure contains genetic material in the typical models?

A. Air bubble

B. Nucleus

C. Glass slide

D. Cell wall

24. Which plant structure mainly provides rigid support?

A. Cell wall

B. Nucleus

C. Eyepiece

D. Coverslip

25. What does the central vacuole contain?

A. A microscope bulb

B. Fuel gas

C. A metal spring

D. Cell sap

26. Which structure carries out photosynthesis?

A. Cell wall alone

B. Stage clip

C. Chloroplast

D. Coverslip

27. Where is the membrane relative to the plant cell wall?

A. Only in air

B. Inside the wall

C. Always outside the wall

D. Inside the eyepiece

28. Which is a careful observation claim?

A. Only describe structures actually visible

B. Draw every structure from memory as observed

C. Assume all dots are nuclei

D. Invent details to match a diagram

29. State one function of the cell membrane.

30. State one role of the cytoplasm.

31. State one role of the nucleus.

32. State one function of the plant cell wall.

33. State a role of the large central vacuole.

34. What is the function of a chloroplast?

35. Why must membrane and wall labels not be swapped?

36. Does cytoplasm mean the cell is empty?

37. Why should a tiny visible dot not automatically be labelled a nucleus?

38. Can a light microscope show every detail of a complex cell diagram?

39. Why is support important for plant cells?

40. Give a structure-and-function pair for a typical animal cell.

Compare fairly

41. Which is shared by the illustrated plant and animal cells?

A. Plant cell wall

B. Chloroplasts

C. A glass wall

D. Cytoplasm

42. Which is not a universal test of cell type?

A. Specimen identity

B. Evidence from preparation

C. Shape alone

D. A careful comparison of structures

43. An onion-bulb cell without chloroplasts is necessarily what?

A. A chemical element

B. Still potentially a plant cell

C. Automatically an animal cell

D. Not living matter

44. Which is a suitable comparison method?

A. Record similarities and differences

B. List differences only from memory

C. Guess from colour alone

D. Invent structures

45. The typical animal model lacks which structure?

A. Cell membrane

B. Cytoplasm

C. Nucleus

D. Plant cell wall

46. A large central vacuole is typical of which illustrated model?

A. Microscope eyepiece

B. Fire triangle

C. Plant cell

D. Both in exactly the same way

47. Which statement avoids overgeneralisation?

A. All animal cells are circular

B. Specialised cells can differ from typical diagrams

C. All cells are identical

D. All plant cells are green

48. Why may a plant diagram show a rectangular outline?

A. To emphasise a supporting wall

B. Because every real plant cell is a perfect rectangle

C. Because plants contain glass

D. Because magnification creates walls

49. List three shared features of the typical models.

50. List two features of the photosynthetic plant model absent from the animal model.

51. Why is green colour alone not a reliable cell test?

52. Why can onion cells lack chloroplasts?

53. What should a comparison table separate?

54. Can a plant cell have both a membrane and wall?

55. What is wrong with saying animal cells have no boundary?

56. Why should two microscope observations be compared at a known magnification?

57. Does a rounded outline prove a specimen is animal tissue?

58. How should an absent visible feature be recorded?

59. Give a fair comparison involving vacuoles.

60. Why is a typical model useful despite exceptions?

Virtual microscope

61. How is total microscope magnification found?

A. Add the two values

B. Subtract the larger value

C. Use only the slide colour

D. Multiply eyepiece by objective magnification

62. A 10-times eyepiece and 10-times objective give what total?

A. 10 times

B. 1 time

C. 100 times

D. 20 times

63. A 10-times eyepiece and 40-times objective give what total?

A. 4 times

B. 400 times

C. 50 times

D. 30 times

64. Which term means distinguishing nearby details?

A. Resolution

B. Combustion

C. Fuel

D. Osmosis

65. Does enlarging a blurred image guarantee new detail?

A. Always

B. Only on paper

C. Only at night

D. No

66. Which is a safe microscope action?

A. Use broken glass

B. Carry it by an unsupported cable

C. Follow the taught focusing method

D. Force a lens into the slide

67. Which lens is usually closest to the specimen?

A. The cell nucleus

B. Objective

C. Eyepiece

D. Spectacle frame only

68. Which lens do you look through in an ordinary compound microscope?

A. Eyepiece

B. Cell wall

C. Coverslip only

D. Stage clip

69. Calculate the total for a 10-times eyepiece and 4-times objective.

70. Calculate the total for a 5-times eyepiece and 10-times objective.

71. Calculate the total for a 15-times eyepiece and 10-times objective.

72. What is wrong with calling 10-times plus 40-times a 50-times view?

73. Define magnification simply.

74. Define resolution simply.

75. What should be recorded with a microscope drawing?

76. Why should a lens not be forced towards a slide?

77. Does the virtual enlargement represent a measured real cell size?

78. Why can field of view change when objective magnification changes?

79. What should happen if equipment seems damaged?

80. Why is multiplication important in this topic?

Observe, draw, care

81. A good biological drawing should mainly show what?

A. Every feature imagined

B. Only decorative shading

C. A guessed result

D. Features actually observed

82. What is a temporary slide?

A. A pressure meter

B. A fire extinguisher

C. A short-term viewing preparation

D. A permanent electronic account

83. Who should approve a real specimen and stain?

A. An unrelated game

B. The teacher or responsible supervisor

C. A random social post

D. Any learner guessing

84. Which may be an artefact rather than a cell feature?

A. An air bubble

B. Every labelled diagram part

C. The topic title

D. A correct multiplication

85. Which is a suitable practical safety action?

A. Touch broken glass

B. Taste stain

C. Use mouth samples without approval

D. Handle and dispose of materials as instructed

86. What should accompany labels?

A. Only guesses

B. Invented measurements

C. Clear accurate lines to the intended structures

D. Lines pointing anywhere

87. If a feature is unclear, what is best?

A. Ignore all evidence

B. Record uncertainty and seek guidance

C. Invent a confident label

D. Alter the specimen record

88. Does the virtual focus scene prove a real structure was observed?

A. No

B. Yes, in every learner's sample

C. Only at 400 times

D. Only with a green screen

89. Name two features of a useful biological drawing.

90. Why should drawings avoid invented structures?

91. What is an artefact in microscopy?

92. Why are some stains not suitable for unsupervised use?

93. Why should broken slides be reported?

94. What should you do with biological waste?

95. Why should a drawing have a title?

96. How can an uncertain label be handled honestly?

97. Why should microscope lenses be cared for correctly?

98. Why is a copied textbook diagram not a record of your specimen?

99. What is a safe alternative before a real practical?

100. What should group members share respectfully?

Answer key & explanations

1. D. A basic unit of living things

Cells perform life-related functions.

2. C. Cell membrane

Animal cells do not have plant cell walls.

3. B. Cell wall

The other listed structures are shared in these models.

4. A. Not every plant cell has chloroplasts

Cell type and role affect structures.

5. D. Not necessarily

Colours are an educational key.

6. C. Some living organisms

Life can be unicellular or multicellular.

7. B. Membrane and cytoplasm

These are common features of both models.

8. A. Cells vary and can be specialised

Typical models simplify biological diversity.

9. A basic structural and functional unit of living things.

It is not a unit of chemical element identity.

10. For example, membrane and cytoplasm.

A nucleus is also shown in both typical models.

11. The plant model has a cell wall; the animal model does not.

Chloroplasts are another difference in photosynthetic plant cells.

12. No.

For example, many non-photosynthetic plant cells lack them.

13. Made of one cell.

Some organisms carry out life processes within one cell.

14. Made of many cells.

Different cells can have specialised roles.

15. It emphasises structures for learning rather than exact relative sizes.

Labels should acknowledge the simplification.

16. Preparation, staining, focus and cell type affect what is visible.

Do not invent observed details.

17. No.

A cell contains many kinds of molecules and atoms.

18. They can be specialised for different functions.

A typical diagram does not show every possible form.

19. The plant-cell model.

The membrane lies inside that wall.

20. To understand what its colours and symbols represent.

Do not assume colour alone identifies a real structure.

21. D. Cell membrane

The membrane selectively regulates movement.

22. C. Cytoplasm

Cytoplasm supports many cellular processes.

23. B. Nucleus

The nucleus helps regulate activities.

24. A. Cell wall

The wall helps maintain shape.

25. D. Cell sap

The vacuole is a fluid-filled compartment.

26. C. Chloroplast

Photosynthetic cells contain chloroplasts.

27. B. Inside the wall

A plant cell has both structures.

28. A. Only describe structures actually visible

Observation and prior knowledge should be distinguished.

29. Regulating substances entering and leaving.

A membrane is selectively permeable.

30. It is where many cell reactions occur.

It is not an empty space.

31. Containing genetic material and helping control cell activities.

Use the typical-cell context.

32. Supporting the cell and helping maintain shape.

It is distinct from the membrane.

33. Containing cell sap and helping support a turgid plant cell.

Water balance affects its contribution.

34. Photosynthesis.

Not every plant cell has one.

35. They are different structures with different roles.

The wall is outside the membrane in a plant cell.

36. No.

It contains material in which many reactions occur.

37. It may be another feature or an artefact.

Use specimen evidence and teacher guidance.

38. No.

Resolution and preparation limit what is visible.

39. It helps maintain form and contributes to the plant's structure.

Walls and water-filled vacuoles both contribute.

40. For example, membrane: regulates movement.

The function must match a structure actually present.

41. D. Cytoplasm

Cytoplasm is present in both typical models.

42. C. Shape alone

Real shapes vary.

43. B. Still potentially a plant cell

Not all plant cells photosynthesise.

44. A. Record similarities and differences

Use evidence and clear categories.

45. D. Plant cell wall

Do not confuse the wall with the membrane.

46. C. Plant cell

The animal model does not have that same large central compartment.

47. B. Specialised cells can differ from typical diagrams

Typical features are not universal guarantees.

48. A. To emphasise a supporting wall

Diagram shape is a simplification.

49. Membrane, cytoplasm and nucleus.

Specialised cells may differ.

50. Cell wall and chloroplasts.

The context is the illustrated typical models.

51. Model colours are chosen and not every plant cell contains chloroplasts.

Use structure and specimen evidence.

52. Bulb cells are not the same as green photosynthetic leaf cells.

Plant identity does not require every cell to photosynthesise.

53. Shared features and differences.

Clear categories prevent contradictory claims.

54. Yes.

The membrane is inside the wall.

55. They have a cell membrane.

No wall does not mean no boundary.

56. Apparent size depends on magnification.

A bigger image does not necessarily mean a bigger real cell.

57. No.

Shape alone is insufficient.

58. State that it was not observed, rather than inventing it.

Not visible is not always the same as absent.

59. The typical plant model has a large central vacuole; the animal model does not have the same structure.

Avoid claiming all animal cells have no small vesicles.

60. It introduces common structures for comparison.

Its limits must remain clear.

61. D. Multiply eyepiece by objective magnification

Both lenses contribute multiplicatively.

62. C. 100 times

Ten multiplied by ten is one hundred.

63. B. 400 times

Ten multiplied by forty is four hundred.

64. A. Resolution

Resolution is not identical to magnification.

65. D. No

Enlargement cannot recover every missing detail.

66. C. Follow the taught focusing method

Real equipment needs careful handling.

67. B. Objective

The objective is on the specimen side.

68. A. Eyepiece

The eyepiece is the viewing lens.

69. 40 times.

Multiply the two lens magnifications.

70. 50 times.

Five multiplied by ten is fifty.

71. 150 times.

Do not add fifteen and ten.

72. The lens magnifications should be multiplied, giving 400 times.

Addition is the wrong operation.

73. How many times larger the image appears than the object.

It is a size relationship, not a measure of truth.

74. The ability to distinguish nearby details.

A large blurred image may have poor useful resolution.

75. The specimen identity and appropriate magnification information.

Also follow the teacher's drawing requirements.

76. It can damage the lens or break the slide.

Use the approved focusing procedure.

77. No.

It illustrates a magnification relationship.

78. Higher magnification usually shows a smaller part of the specimen.

The virtual model is simplified.

79. Stop and tell the teacher.

Do not repair it unsupervised.

80. It combines the eyepiece and objective magnifications.

The maths supports accurate observation records.

81. D. Features actually observed

Separate observation from prior knowledge.

82. C. A short-term viewing preparation

It is prepared for observation rather than indefinite storage.

83. B. The teacher or responsible supervisor

Biological materials and chemicals need assessment.

84. A. An air bubble

Preparation can produce misleading appearances.

85. D. Handle and dispose of materials as instructed

Follow the approved practical method.

86. C. Clear accurate lines to the intended structures

Communication should be unambiguous.

87. B. Record uncertainty and seek guidance

Honesty is part of science.

88. A. No

It is a teaching model.

89. For example, clear lines and accurate labels.

A suitable title and magnification information also help.

90. They would misrepresent the observation.

Use evidence rather than a preferred result.

91. A feature introduced by preparation or viewing rather than the intended biological structure.

Air bubbles are a common example.

92. They may pose chemical hazards and need an approved method.

The app is not a practical recipe.

93. They can cut people and require safe disposal.

Do not pick up sharp glass with bare hands.

94. Follow the teacher's disposal instructions.

Do not leave it on shared surfaces.

95. It identifies what the drawing represents.

Include the specimen context.

96. State that the feature is unclear and ask for guidance.

Do not present a guess as a confirmed finding.

97. Damage or dirt can impair observations.

Use only the instructed cleaning method.

98. It shows a model, not necessarily what you observed.

Models and observations have different purposes.

99. Use the virtual model and discuss the approved method with the teacher.

Practical skills still need supervised assessment.

100. Their observations, reasons and uncertainties.

Scientific collaboration is not simply copying answers.

Explore responsibly

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